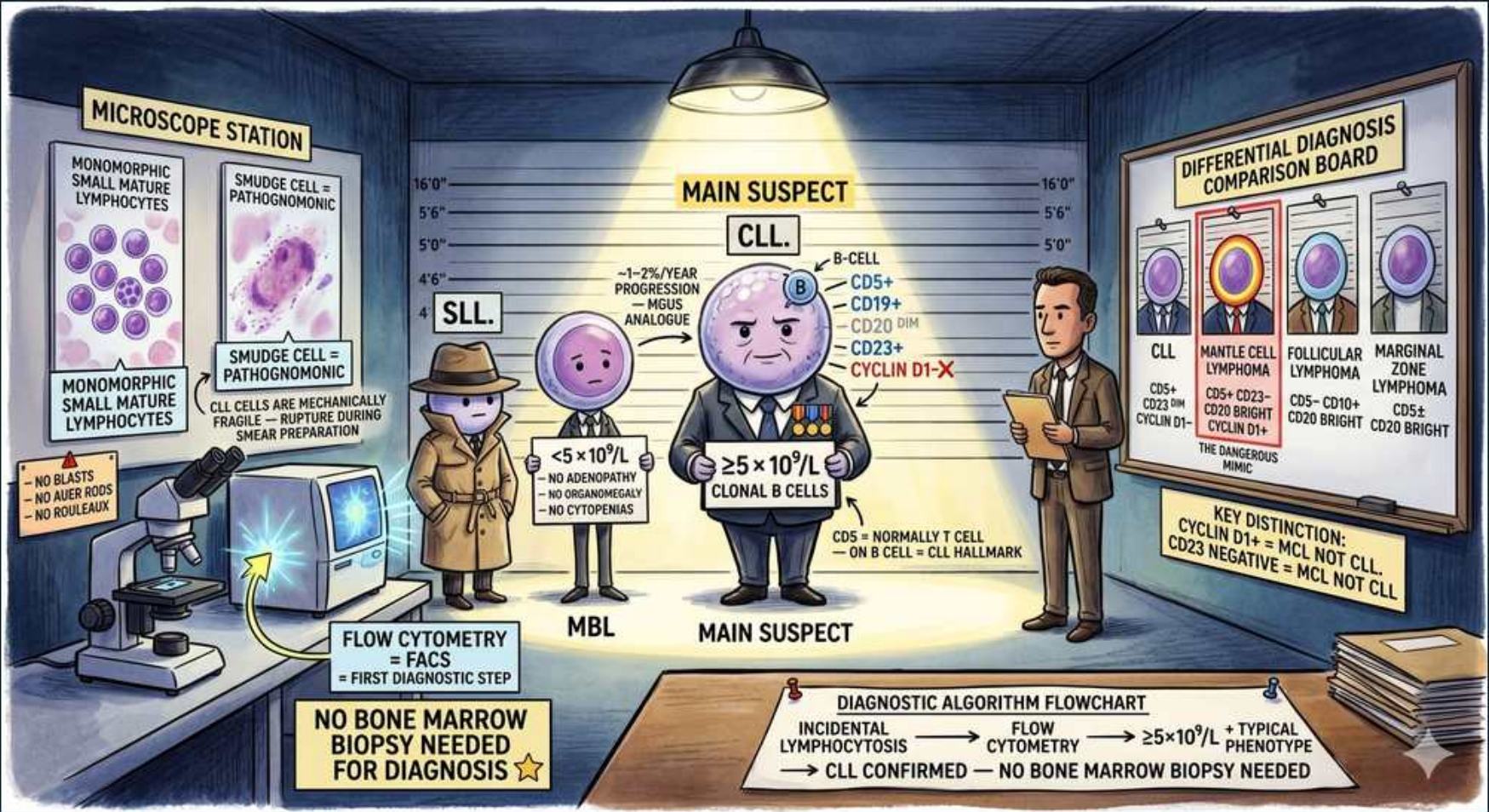


CLL — Diagnosis & Immunophenotype



1. Smudge cells

WHAT YOU SEE

A microscope-station panel of a burst, smeared 'SMUDGE CELL' tagged 'PATHOGNOMONIC' — the fragile CLL cell ruptured during smear prep.

WHAT IT ENCODES

Smudge (basket) cells on the peripheral smear are characteristic of CLL: the fragile, mechanically weak CLL lymphocytes rupture during slide preparation, leaving smeared nuclear debris. Their abundance correlates with CLL (a high smudge-cell fraction favors CLL).

BOARD TRAP

Smudge cells are an artifact of fragile cells, not a distinct cell type — adding albumin to the smear reduces them, confirming they are processing artifacts.

2. Mature lymphocytes

WHAT YOU SEE

A microscope panel of uniform 'MONOMORPHIC SMALL MATURE LYMPHOCYTES' — a sea of identical tiny clumped-chromatin B cells, not blasts.

WHAT IT ENCODES

CLL = monomorphic population of SMALL, MATURE-appearing B lymphocytes with clumped ("soccer-ball") chromatin, scant cytoplasm, and inconspicuous nucleoli — accumulation of long-lived non-dividing cells rather than proliferating blasts.

BOARD TRAP

NOT blasts — there are no Auer rods and no prominent nucleoli (which would suggest acute leukemia or prolymphocytic transformation if >55% prolymphocytes).

3. ≥5×10⁹/L B cells

WHAT YOU SEE

The spotlit lineup 'MAIN SUSPECT' figure labeled '≥5×10⁹/L CLONAL B CELLS' — the blood threshold that fingers true CLL.

WHAT IT ENCODES

CLL diagnosis requires ≥5×10⁹/L clonal B lymphocytes in the peripheral blood sustained for ≥3 months, with clonality confirmed by flow (light-chain restriction, kappa OR lambda). This blood threshold is what separates CLL from MBL and SLL.

BOARD TRAP

It is the clonal B-cell count, not the total lymphocyte count, that must be ≥5×10⁹/L.

4. CD5+

WHAT YOU SEE

A badge 'CD5+' on the suspect with the note 'normally a T cell marker — on B cell = CLL hallmark' — the aberrant T-cell antigen on a B cell.

WHAT IT ENCODES

CLL B cells aberrantly co-express CD5 (normally a T-cell antigen) alongside B-cell markers. Full CLL immunophenotype: CD5+, CD19+, CD23+, CD20 dim, CD200+, and DIM surface immunoglobulin with light-chain restriction.

BOARD TRAP

CD5 is also positive in mantle cell lymphoma — CD5+ alone does not equal CLL; you must add CD23 and cyclin D1 to separate them.

5. CD23+

WHAT YOU SEE

A 'CD23+' badge pinned on the MAIN SUSPECT board — the marker whose presence supports CLL over its mimics.

WHAT IT ENCODES

CLL is CD23 POSITIVE — a defining marker that, combined with CD5+ and dim surface Ig, supports CLL. The classic CLL panel is CD5+ / CD19+ / CD23+ / CD20-dim / FMC7-negative / dim slg.

BOARD TRAP

CD23 NEGATIVE points to MANTLE CELL lymphoma, not CLL (the single most tested discriminator).

6. Cyclin D1 negative

WHAT YOU SEE

'CYCLIN D1' on the suspect with a red X struck through it — CLL is cyclin-D1 negative, ruling out the t(11;14) of mantle cell.

WHAT IT ENCODES

CLL is cyclin D1 NEGATIVE and SOX11 negative — it lacks the t(11;14) translocation. This is the key confirmatory step when CD5+ B-cell disease is found.

BOARD TRAP

Cyclin D1 POSITIVE (with t(11;14) and SOX11+) = MANTLE CELL lymphoma, NOT CLL. CD5+/CD23-/cyclin D1+ = mantle cell; CD5+/CD23+/cyclin D1- = CLL.

7. Flow cytometry first

WHAT YOU SEE

The flowchart arrow 'FLOW CYTOMETRY = FACS = FIRST DIAGNOSTIC STEP' with a glowing analyzer — blood immunophenotyping clinches the diagnosis.

WHAT IT ENCODES

Peripheral-blood FLOW CYTOMETRY (immunophenotyping) is the first and definitive diagnostic step — it demonstrates the clonal CD5+CD19+CD23+ dim-slg population, avoiding the need for tissue or marrow in most cases.

8. No marrow biopsy

WHAT YOU SEE

A gold-starred sign 'NO BONE MARROW BIOPSY NEEDED FOR DIAGNOSIS' — blood flow alone suffices; the star marks the high-yield rule.

WHAT IT ENCODES

Bone marrow biopsy is NOT required to diagnose CLL — peripheral blood flow suffices. Marrow is reserved for clarifying the cause of cytopenias (autoimmune vs marrow infiltration) or pre-treatment assessment, not for diagnosis.

BOARD TRAP

Don't order a marrow biopsy just to confirm CLL — the diagnosis is made on blood flow cytometry.

9. MBL <5×10³

WHAT YOU SEE

A smaller lineup figure 'MBL' tagged '<5×10³/L — no adenopathy / no organomegaly / no cytopenias' — the sub-threshold precursor state.

WHAT IT ENCODES

Monoclonal B-cell lymphocytosis (MBL): clonal CLL-phenotype B cells <5×10³/L WITHOUT lymphadenopathy, organomegaly, cytopenias, or disease symptoms — a precursor state requiring only observation.

BOARD TRAP

MBL is defined by absence of disease features AND clonal B cells <5×10³/L; if any node/organomegaly/cytopenia is present it is CLL/SLL, not MBL.

10. SLL = tissue

WHAT YOU SEE

A trench-coated 'SLL' figure standing apart from the blood lineup — the same suspect hiding in lymph node/tissue instead of blood.

WHAT IT ENCODES

Small lymphocytic lymphoma (SLL) is the SAME disease as CLL but presents in lymph nodes/tissue rather than blood: lymphadenopathy/organomegaly with <5×10³/L blood clonal B cells. Diagnosis requires lymph-node biopsy.

BOARD TRAP

CLL and SLL are one entity (CLL/SLL) differing only in compartment — same immunophenotype, staged/treated similarly.

11. 1-2%/yr progression

WHAT YOU SEE

A caption '~1-2%/year progression — MGUS analogue' beside MBL — the slow drift of MBL toward treatment-requiring CLL.

WHAT IT ENCODES

High-count MBL progresses to treatment-requiring CLL at roughly 1–2% per year, justifying periodic monitoring without therapy.

12. Mantle cell DDx

WHAT YOU SEE

The DDx comparison board column 'MANTLE CELL LYMPHOMA — CD5+ CD23- cyclin D1+', boxed in red as 'THE DANGEROUS MIMIC'.

WHAT IT ENCODES

Mantle cell lymphoma is the chief mimic: CD5+ (like CLL) but CD23 NEGATIVE, cyclin D1 POSITIVE, SOX11+, with t(11;14)(IGH-CCND1). Distinguishing it from CLL is the highest-yield immunophenotyping point.

BOARD TRAP

A CD5+ B-cell lymphoproliferative disorder is NOT automatically CLL — check CD23 and cyclin D1 to exclude mantle cell.

13. Follicular DDx

WHAT YOU SEE

DDx board columns 'FOLLICULAR — CD5- CD10+' and 'MARGINAL ZONE — CD5±' — the CD5-negative lymphomas that separate cleanly from CLL.

WHAT IT ENCODES

Follicular lymphoma: CD5 NEGATIVE, CD10 POSITIVE (germinal-center marker), BCL2+ with t(14;18). Marginal zone lymphoma is CD5–/CD10–. These CD5-negative entities separate cleanly from CLL.

BOARD TRAP

CD10 positivity points to a germinal-center lymphoma (follicular/Burkitt/DLBCL-GCB), not CLL — CLL is CD10 negative.

14. Incidental lymphocytosis

WHAT YOU SEE

The bottom 'DIAGNOSTIC ALGORITHM FLOWCHART': incidental lymphocytosis → flow cytometry → ≥5×10³/L + typical phenotype → CLL confirmed.

WHAT IT ENCODES

Most CLL is discovered incidentally as asymptomatic absolute lymphocytosis on a routine CBC; the diagnostic algorithm is incidental lymphocytosis → flow cytometry → clonal B cells ≥5×10³/L + typical phenotype → CLL confirmed without marrow biopsy.